Feature Description 588/221 04-LZA 701 6014/1 Uen S32C

LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

Contents

LTE Decoupled Downlink and Uplink PDCCH Link Adaptation Overview

Dependencies of LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

Feature Operation LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

Network Impact of LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

Parameters for LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

Performance of LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

Activate LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

Deactivate LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

Engineering Guidelines for LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

Appendix: Feature Change History

- Appendix: 24.Q4.0: Improvement for PDCCH Outer Loop Adjustment
- Appendix: 25.Q1.0: Enhanced LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

1 LTE Decoupled Downlink and Uplink PDCCH Link Adaptation Overview

> This feature enables decoupling the PDCCH outer loop adjustment and PDCCH link adaptation into uplink and downlink to avoid PDCCH congestion.

Feature Name	LTE Decoupled Downlink and Uplink PDCCH Link Adaptation
Feature Identity	FAJ 121 5716
Value Package Name	High Load Handling
Value Package Identity	FAJ 801 0415
Node Type	Baseband Radio Node
Access Type	LTE
Licensing	License-controlled feature. One license is required for each node.

Summary

This feature enables decoupling the PDCCH outer loop link adaptation scheme into uplink and downlink processes.

With decoupling, PDCCH outer loop adjustment is improved if DTX is caused by PUSCH or PUCCH. Decoupling the uplink and downlink PDCCH outer loop link adaptation helps to avoid unnecessary reduction of PDCCH uplink or downlink SINR.

The feature can also replace uplink PDCCH outer loop link adaptation with downlink PDCCH outer loop link adaptation if the DTX difference exceeds the threshold set by the `EUtranCellFDD.pdcchUlDtxWorseThres` or the `EUtranCellTDD.pdcchUlDtxWorseThres` attribute.

If the PDCCH downlink DTX rate is higher than the PDCCH uplink DTX rate, downlink DTX can be limited for decreasing downlink PDCCH SINR.

This feature has the following benefits:

- PDCCH outer loop link adaptation is improved.
- Interference of the uplink and downlink PDCCH is avoided.
- The CCE aggregation level is optimized.
- PDCCH congestion might be prevented.

Additional Information

For information on the preferred state and parameter settings of this feature, see RAN Parameter Recommendations Lists.

2 Dependencies of LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

> This feature is related to features associated with PDCCH optimization and uplink DTX increase.

Table 1 Feature Dependencies

Feature	Relationship	Relationship Scope	Description
Enhanced PDCCH Link Adaptation (FAJ 121 3051)	Prerequisite	Cell-level prerequisite	The Enhanced PDCCH Link Adaptation feature must be activated for the LTE Decoupled Downlink and Uplink PDCCH Link Adaptation feature to be operational.
Optimized PDCCH (FAJ 121 5344)	Related		If both features are activated simultaneously, the following outcomes are expected: – PDCCH congestion can be reduced if there is high cell load. – The Optimized PDCCH feature might be triggered less often, as it does not require such a high target BLER to reduce congestion. If the Optimized PDCCH feature is activated, the PDCCH congestion measurement period used in the LTE Decoupled Downlink and Uplink PDCCH Link Adaptation feature uses the value configured in the `EUtranCellFDD.pdcchCongestMeasurePeriod` or `EUtranCellTDD.pdcchCongestMeasurePeriod` attribute.
Uplink Scheduling Control for Out-of-Coverage UEs (FAJ 121 4524)	Related		If both features are activated simultaneously, the uplink DTX is increased. This increase leads to an increased number of UEs in out-of-coverage state in the cell. Therefore, the counts of the `EUtranCellFDD.pmOocStateEntries` and `EUtranCellTDD.pmOocStateEntries` PM counters are increased.

Hardware

No specific hardware requirements.

Limitations

No known limitations.

Network Requirements

It is recommended to use ENM version 24.1 or later to support the MOM changes introduced by this feature. If the Release Independence Manager application is used, the MOM changes are supported on earlier ENM versions also.

Related concepts
Engineering Guidelines for LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

3 Feature Operation LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

> This feature splits the PDCCH link adaptation scheme into uplink and downlink.

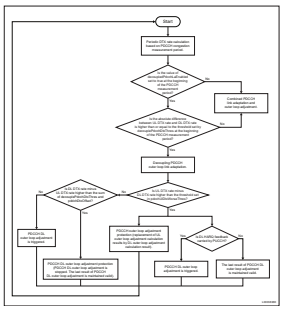
PDCCH outer loop link adaptation based on combined uplink and downlink HARQ measurement leads to the incorrect adjustment of PDCCH if the difference between uplink and downlink DTX rate is obvious. By separating uplink and downlink PDCCH outer loop link adaptation, PDCCH link adjustment can be more accurate.

To enable decoupling of uplink and downlink PDCCH link adaptation, the `EUtranCellFDD.decoupledPdcchLaEnabled` or the `EUtranCellTDD.decoupledPdcchLaEnabled` attribute must be set to `true`.

The feature uses the following attributes as threshold for the absolute value of uplink DTX rate minus downlink DTX rate:

- `EUtranCellFDD.decouplePdcchDtxThres`
- `EUtranCellTDD.decouplePdcchDtxThres`

A higher threshold value means that decoupling is less likely to be triggered.


Start
Periodic DTX rate calculation based on congestion measurement period
Is the value of the EUtranCellFDD.decoupledPdcchLaEnabled or the EUtranCellTDD.decoupledPdcchLaEnabled attribute set to true?
No
Combined PDCCH uplink and downlink outer loop adjustment
Yes
Is the absolute difference between UL DTX rate and DL DTX rate is higher than or equal to the threshold set by the EUtranCellFDD.decouplePdcchDtxThres or the EUtranCellTDD.decouplePdcchDtxThres attribute?
No
Yes
Decoupling PDCCH uplink link adaptation
Is UL DTX rate minus DL DTX rate higher than the EUtranCellFDD.decouplePdcchDtxThres or the EUtranCellTDD.decouplePdcchDtxThres?
No
Is UL DTX rate minus DL DTX rate higher than the EUtranCellFDD.decouplePdcchDtxThres or the EUtranCellTDD.decouplePdcchDtxThres?
No
Yes
Yes
PDCCH DL outer loop adjustment is triggered
PDCCH UL outer loop adjustment is triggered
PDCCH UL outer loop adjustment is triggered
PDCCH U

Otherwise, uplink and downlink PDCCH outer loop link adaptation is combined.

4. When decoupling PDCCH uplink and downlink outer loop link adaptation is triggered, the feature checks if the difference that refers to the downlink DTX rate subtracted from the uplink DTX rate, exceeds a preset threshold. The threshold is defined by configuring the value of the `EUtranCellFDD.pdcchUlDtxWorseThres` or the `EUtranCellTDD.pdcchUlDtxWorseThres` attribute.

5. If the preset threshold is exceeded, the feature leads to one of the following outcomes:

   – The uplink PDCCH outer loop link adaptation is replaced by the downlink PDCCH outer loop link adaptation. As a result of the replacement, a lower CCE aggregation level is selected for the uplink that helps to avoid PDCCH congestion.

   – PDCCH downlink outer loop adjustment is implemented if downlink HARQ is carried by PUCCH. Therefore, the feature helps avoid the selection of a higher CCE level if the eNodeB does not receive downlink HARQ feedback because of poor PUSCH.

     If downlink HARQ feedback is carried by PUSCH, downlink outer loop adjustment is not triggered. In this case, the last result of PDCCH downlink outer loop adjustment calculation is maintained.

6. The feature checks if the difference that refers to the uplink DTX rate subtracted from the downlink DTX rate is higher than the sum of the values of the following attributes:

   – `EUtranCellFDD.decouplePdcchDtxThres` and `EUtranCellFDD.pdcchDlOlaDtxOffset`

   – `EUtranCellTDD.decouplePdcchDtxThres` and `EUtranCellTDD.pdcchDlOlaDtxOffset`

   If the downlink DTX rate minus uplink DTX rate is higher than the sum of the attribute values, downlink PDCCH outer loop adjustment triggered by downlink DTX is stopped. The last result of PDCCH downlink outer loop adjustment calculation is maintained.

For more information, see Engineering Guidelines.

Related concepts
Engineering Guidelines for LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

4 Network Impact of LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

> This feature can help to avoid PDCCH outer loop link adaptation mutual influence between uplink and downlink and might relieve PDCCH congestion, especially in high load traffic.

Capacity and Performance

If uplink and downlink payload volume is less restricted by PDCCH exhaustion, improved uplink and downlink cell throughput is expected in high load conditions. As the primary benefit of the feature, the average CCE aggregation level might decrease.

The feature affects the number of scheduled entities for a TTI in the following ways:

- The number of scheduled entities for a TTI might increase as a result of reduced restriction by CCE exhaustion.
- The average PDCCH CCE use decreases under the same number of scheduled entities for a TTI.
- If more scheduled entities for a TTI are scheduled, CCE use might increase.

With decoupling, PDCCH uses less CCE resource for uplink scheduling so the UE might fail to receive uplink scheduling requests.

Interfaces

No impact.

Other Network Elements

No impact.

5 Parameters for LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

> This feature introduces attributes to enable the separation and optimization of downlink and uplink PDCCH link adaptation.

Table 2 Parameters

Parameter	Type	Description
EUtranCellFDD.decouplePdcchDtxThres	Introduced	See MOM description.
EUtranCellTDD.decouplePdcchDtxThres	Introduced	See MOM description.
EUtranCellFDD.decoupledPdcchLaEnabled	Introduced	See MOM description.
EUtranCellTDD.decoupledPdcchLaEnabled	Introduced	See MOM description.
EUtranCellFDD.pdcchDlOlaDtxOffset	Introduced	See MOM description.
EUtranCellTDD.pdcchDlOlaDtxOffset	Introduced	See MOM description.
EUtranCellFDD.pdcchUlDtxWorseThres	Introduced	See MOM description.
EUtranCellTDD.pdcchUlDtxWorseThres	Introduced	See MOM description.
EUtranCellFDD.pdcchCongestMeasurePeriod	Affecting	The feature uses the configured value of this attribute for PDCCH congestion measurement period, if the Optimized PDCCH feature is activated.
EUtranCellTDD.pdcchCongestMeasurePeriod	Affecting	The feature uses the configured value of this attribute for PDCCH congestion measurement period, if the Optimized PDCCH feature is activated.

6 Performance of LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

> This feature is associated with throughput-related KPIs. The feaure introduces counters and events related to measuring the difference between PDCCH uplink DTX ratio and PDCCH downlink DTX ratio.

Key Performance Indicators

Table 3 Key Performance Indicators

Key Performance Indicator	Description
Average DL MAC Cell Throughput	This KPI is expected to increase, because better cell throughput is expected in high load conditions. The increase is possible as PDCCH exhaustion affects downlink and uplink payload volume less.
Average DL PDCP Cell Throughput	This KPI is expected to increase, as better cell throughput is expected in high load conditions. The increase is possible as PDCCH exhaustion affects downlink and uplink payload volume less.
Average UL PDCP Cell Throughput	This KPI is expected to increase, as better cell throughput is expected in high load conditions. The increase is possible as PDCCH exhaustion affects downlink and uplink payload volume less.
Average UL MAC Cell Throughput	This KPI is expected to increase, as better cell throughput is expected in high load conditions. The increase is possible as PDCCH exhaustion affects downlink and uplink payload volume less.

For more information on KPIs, see Key Performance Indicators.

Table 4 Performance Indicators

Performance Indicator	Formula	Description
Average CCE Aggregation Level	`AvgPdcchCceAggrLevel = 100 × (pmPdcchCceAggregationDistr [0] × 1 + pmPdcchCceAggregationDistr [1] × 2+ pmPdcchCceAggregationDistr [2] × 4 + pmPdcchCceAggregationDistr [3] × 8) / Σ pmPdcchCceAggregationDistr [i]`	The trend of CCE aggregation distribution might shift from high CCE levels to low CCE levels. Average CCE aggregation level is decreased.
Average PDCCH CCE Utilization	`AvgPdcchCceUtil = Σ pmPdcchCceUtil [i] × (i × 5+2.5) / Σ pmPdcchCceUtil [i]; i = 0,1,….,19`	Downlink CCE use might decrease while uplink CCE use might increase. The higher the number of Scheduling Entities per TTI, the higher the CCE use.
Scheduling Entities per TTI DL	`SE per TTI DL = pmSchedActivityUeDl / pmSchedTtiCellDl`	Scheduling Entities per TTI DL PI might increase as the restriction by CCE exhaustion lessens.
Scheduling Entities per TTI UL	`SE per TTI UL = pmSchedActivityUeUl / pmSchedTtiCellUl`	Scheduling Entities per TTI UL PI might increase as the restriction by CCE exhaustion lessens.

Performance Indicator	Formula	Description
PDCCH Packet Error Rate	PDCCH Packet Error Rate [%] = 100 × ((pmDlAssigsTrans + pmUlGrantsTrans - pmUlGrantsTransNoAck - pmDlAssigsWithDetectedHarqAck - pmUlGrantsWithDetectedPusch - pmMacSkipUlDtxDynamic) / (pmDlAssigsTrans + pmUlGrantsTrans - pmUlGrantsTransNoAck - pmMacSkipUlDtxDynamic))	PDCCH Packet Error Rate PI might increase.

PM Counters

This feature introduces the following PM counters:

- EUtranCellFDD.pmDlSinrPdcchDistr
- EUtranCellFDD.pmPdcchCceAggregationDlDistr
- EUtranCellFDD.pmPdcchCceAggregationUlDistr
- EUtranCellFDD.pmPdcchDlDtxWorseDistr
- EUtranCellFDD.pmPdcchUlDtxWorseDistr
- EUtranCellFDD.pmUlSinrPdcchDistr
- EUtranCellTDD.pmDlSinrPdcchDistr
- EUtranCellTDD.pmPdcchCceAggregationDlDistr
- EUtranCellTDD.pmPdcchCceAggregationUlDistr
- EUtranCellTDD.pmPdcchDlDtxWorseDistr
- EUtranCellTDD.pmPdcchUlDtxWorseDistr
- EUtranCellTDD.pmUlSinrPdcchDistr

In addition to the introduced PM counters, the feature can be observed through the following PM counters:

- EUtranCellFDD.pmOocStateEntries
- EUtranCellFDD.pmPdcchCongestDistr
- EUtranCellFDD.pmPucchFailSr
- EUtranCellTDD.pmOocStateEntries
- EUtranCellTDD.pmPdcchCongestDistr
- EUtranCellTDD.pmPucchFailSr

Events

Table 5 Events Associated with the Feature

Event	Event Parameter
LOCAL_PER_BB_CELL5_EVENT	EVENT_PARAM_PDCCH_DL_DTX_WORSE_DISTR_XXX
LOCAL_PER_BB_CELL5_EVENT	EVENT_PARAM_PDCCH_UL_DTX_WORSE_DISTR_XXX
LOCAL_PER_BB_CELL5_EVENT	EVENT_PARAM_PDCCH_CCE_AGGREGATION_LEVEL_DL_DISTR_XXX
LOCAL_PER_BB_CELL5_EVENT	EVENT_PARAM_PDCCH_CCE_AGGREGATION_LEVEL_UL_DISTR_XXX

Event	Event Parameter
LOCAL_PER_BB_UE_ON_CELL_LEVEL_EVENT	EVENT_PARAM_SINR_MEAS_PDCCH_DL _XXX EVENT_PARAM_SINR_MEAS_PDCCH_DL _XXX_XL
LOCAL_PER_BB_UE_ON_CELL_LEVEL_EVENT	EVENT_PARAM_SINR_MEAS_PDCCH_UL_XXX EVENT_PARAM_SINR_MEAS_PDCCH_UL _XXX_XL

For a full list with detailed information about counters, see the list files in the List Files library folder.

7 Activate LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

> To use the feature, it must be activated following the usual feature activation procedure.

Prerequisites

- The license key is installed in the node.
- The prerequisite feature is activated.
- CCTR is active for at least one week before this procedure to collect enough troubleshooting data.

Steps

1. Lock the cell.
2. Set the `FeatureState.featureState` attribute to `ACTIVATED` in the `FeatureState=CXC4012732` MO instance.
3. Set the `EUtranCellFDD.decoupledPdcchLaEnabled` and `EUtranCellTDD.decoupledPdcchLaEnabled` attributes to `true` in their respective deployments.
4. Unlock the cell.

After This Task

Keep CCTR active for at least one week after this procedure for continued collection of troubleshooting data.

8 Deactivate LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

> If the feature is no longer needed, it can be deactivated following the usual feature deactivation procedure. It must also be deactivated before the activation of any conflicting feature.

Prerequisites

- CCTR is active for at least one week before this procedure to collect enough troubleshooting data.

Steps

1. Lock the cell.
2. Set the `FeatureState.featureState` attribute to `DEACTIVATED` in the `FeatureState=CXC4012732` MO instance.
3. Set the `EUtranCellFDD.decoupledPdcchLaEnabled` and `EUtranCellTDD.decoupledPdcchLaEnabled` attributes to `false` in their respective deployments.
4. Unlock the cell.

After This Task

Keep CCTR active for at least one week after this procedure for continued collection of troubleshooting data.

9 Engineering Guidelines for LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

> Triggering decoupling depends on the values of the threshold-related attributes.

The operator can obtain the number of samples of DTX rate difference distribution through the following counters on particular bins:

- `EUtranCellFDD.pmPdcchUlDtxWorseDistr`
- `EUtranCellTDD.pmPdcchUlDtxWorseDistr`
- `EUtranCellFDD.pmPdcchDlDtxWorseDistr`
- `EUtranCellTDD.pmPdcchDlDtxWorseDistr`

Each bin represents the range of difference between uplink DTX rate and downlink DTX rate and the amount of pegging in each bin represents how frequent the given difference took place during each ROP.

The value of the `EUtranCellFDD.decouplePdcchDtxThres` or the `EUtranCellTDD.decouplePdcchDtxThres` attribute affects how frequently decoupling uplink and downlink PDCCH link adaptation is triggered within the observation period.

The values of the following attributes affects how frequently uplink and downlink PDCCH outer loop adjustment protection is triggered within the observation period:

- `EUtranCellFDD.decouplePdcchDtxThres`
- `EUtranCellFDD.pdcchDlOlaDtxOffset`
- `EUtranCellFDD.pdcchUlDtxWorseThres`

or

- `EUtranCellTDD.decouplePdcchDtxThres`
- `EUtranCellTDD.pdcchDlOlaDtxOffset`
- `EUtranCellTDD.pdcchUlDtxWorseThres`

The lower these thresholds are set, the more frequently decoupling uplink and downlink PDCCH link adaptation and PDCCH outer loop adjustment protection are triggered.

Applicable Scenario with Decoupling

The percentage value of the PM counters represents sum of all samples collected on the considered bin indexes with respect to the maximum possible samples. The higher proportion of higher bins is, the more obvious the difference and the greater gain are.

Table 6 Statistics in a 15-minutes ROP

	Bin 0	Bin 1	Bin 2	Bin 3	Bin 4	Bin 5	Bin 6	Bin 7	Total Number of Samples
pmPdcchUlDtxWorseDistr	43	25	138	395	34	2	0	0	635
Difference Rate	6.77%	2.78%	15.33%	43.89%	5.35%	0.22%	0.00%	0.00%	
pmPdcchDlDtxWorseDistr	200	65	0	0	0	0	0	0	265
Difference Rate	22.22%	7.22%	0	0	0	0	0	0	

	Bin 0	Bin 1	Bin 2	Bin 3	Bin 4	Bin 5	Bin 6	Bin 7	Total Number of Samples
Total Small Difference of Bin 0 and Bin 1	38.99%		N/A						
Obvious Difference of Bin 2 ~ Bin 7	N/A		64.80%						

Disabling Conditional Decoupling

Checking the absolute value of DTX rate difference is a condition involved in decision making whether to trigger decoupling.

Conditional decoupling is disabled through the following settings:

- The value of the `EUtranCellFDD.decouplePdcchDtxThres` or the `EUtranCellTDD.decouplePdcchDtxThres` attribute is set to zero.
- The sum of the following attributes is greater than or equal to `100`:
  - `EUtranCellFDD.pdcchDlOlaDtxOffset` and `EUtranCellFDD.decouplePdcchDtxThres`
  - `EUtranCellTDD.pdcchDlOlaDtxOffset` and `EUtranCellTDD.decouplePdcchDtxThres`

Note: The feature is recommended to use with conditional decoupling.

Appendix A: Feature Change History

> This section lists changes that affected this feature and the impact it had on the network.

Appendix A.a: 24.Q4.0: Improvement for PDCCH Outer Loop Adjustment

> The default value of the `EUtranCellFDD.pdcchUlDtxWorseThres` and the `EUtranCellTDD.pdcchUlDtxWorseThres` attributes are changed from 10 to 5.

Feature Name	LTE Decoupled Downlink and Uplink PDCCH Link Adaptation
Feature Identity	FAJ 121 5716
Value Package Name	High Load Handling
Value Package Identity	FAJ 801 0415
Node Type	Baseband Radio Node
Access Type	LTE
Impact at Upgrade	Mandatory

Summary and Benefits

The decreased value of the `EUtranCellFDD.pdcchUlDtxWorseThres` or the `EUtranCellTDD.pdcchUlDtxWorseThres` attribute can trigger more uplink PDCCH outer loop adjustments replaced by downlink PDCCH outer loop adjustments when the uplink DTX ratio is worse than the downlink DTX ratio. This replacement leads to optimized downlink and uplink outer loop adjustment.

The decreased value of the `EUtranCellFDD.pdcchUlDtxWorseThres` and the `EUtranCellTDD.pdcchUlDtxWorseThres` attributes can also result in triggering more PDCCH downlink outer loop adjustments if downlink HARQ is carried by PUCCH. This way, a higher CCE level is not selected if the eNodeB does not receive downlink HARQ feedback because of poor PUSCH.

Capacity and Performance

The feature enhancement might affect the following KPIs:

- Average DL MAC Cell Throughput
- Average DL PDCP Cell Throughput
- Average UL MAC Cell Throughput
- Average UL PDCP Cell Throughput

The feature enhancement might affect the following performance indicators:

- Average CCE Aggregation Level
- Average PDCCH CCE Utilization
- Scheduling Entities per TTI DL
- Scheduling Entities per TTI UL

The feature enhancement might affect the following PM counters:

- `EUtranCellFDD.pmDlSinrPdcchDistr`
- `EUtranCellFDD.pmPdcchCceAggregationDlDistr`
- `EUtranCellFDD.pmPdcchCceAggregationUlDistr`
- `EUtranCellFDD.pmPdcchDlDtxWorseDistr`

- EUtranCellFDD.pmPdcchUlDtxWorseDistr
- EUtranCellFDD.pmUlSinrPdcchDistr
- EUtranCellTDD.pmDlSinrPdcchDistr
- EUtranCellTDD.pmPdcchCceAggregationDlDistr
- EUtranCellTDD.pmPdcchCceAggregationUlDistr
- EUtranCellTDD.pmPdcchDlDtxWorseDistr
- EUtranCellTDD.pmPdcchUlDtxWorseDistr
- EUtranCellTDD.pmUlSinrPdcchDistr

Operation

This feature enhancement decreases the default value of the EUtranCellFDD.pdcchUlDtxWorseThres attribute and the EUtranCellTDD.pdcchUlDtxWorseThres attribute to 5.

Interfaces

No impact.

Hardware

No specific hardware requirements.

Other Network Elements

It is recommended to use ENM version 24.4 or later to support the MOM changes introduced by this feature. If the Release Independence Manager application is used, the MOM changes are supported on earlier ENM versions also.

Related Information
LTE Decoupled Downlink and Uplink PDCCH Link Adaptation
Enhanced PDCCH Link Adaptation

Appendix A.b: 25.Q1.0: Enhanced LTE Decoupled Downlink and Uplink PDCCH Link Adaptation

> This feature enhancement introduces conditions for decoupling PDCCH link adaptation and outer loop adjustment to optimize CCE consumption.

Feature Name	LTE Decoupled Downlink and Uplink PDCCH Link Adaptation
Feature Identity	FAJ 121 5716
Value Package Name	High Load Handling
Value Package Identity	FAJ 801 0415
Node Type	Baseband Radio Node
Access Type	LTE
Impact at Upgrade	Optional

Summary and Benefits

This feature enhancement provides further options for decoupling between uplink and downlink PDCCH link adaptation and outer loop adjustment through checking the absolute value of DTX rate difference between uplink DTX rate and downlink DTX rate.

At the beginning of each PDDCH measurement period, the feature enhancement compares the absolute value of DTX rate difference to a threshold. If the absolute value of DTX rate difference is greater than or equal to the threshold,

decoupling PDCCH uplink and downlink PDCCH link adaptation and outer loop adjustment is triggered. The threshold is set by the value of the `EUtranCellFDD.decouplePdcchDtxThres` or the `EUtranCellTDD.decouplePdcchDtxThres` attribute.

If the PDCCH downlink DTX rate is higher than the PDCCH uplink DTX rate, downlink DTX can be limited for decreasing downlink PDCCH SINR.

Compared to unconditional decoupling, conditional decoupling helps to prevent the impact of increased average CCE aggregation level. This prevention is especially important if the difference between the uplink DTX rate and the downlink DTX rate is small at feature activation.

Capacity and Performance

This feature enhancement might affect the following KPIs:

- Average DL MAC Cell Throughput
- Average DL PDCP Cell Throughput
- Average UL MAC Cell Throughput
- Average UL PDCP Cell Throughput

This feature enhancement might affect the following Performance Indicators:

- Average CCE Aggregation Level
- Average PDCCH CCE Utilization
- Scheduling Entities per TTI DL
- Scheduling Entities per TTI UL

Also, this feature enhancement might increase the PDCCH Packet Error Rate Performance Indicator.

This feature enhancement might affect the following PM counters:

- `EUtranCellFDD.pmDlSinrPdcchDistr`
- `EUtranCellTDD.pmDlSinrPdcchDistr`
- `EUtranCellFDD.pmOocStateEntries`
- `EUtranCellTDD.pmOocStateEntries`
- `EUtranCellFDD.pmPdcchCceAggregationDlDistr`
- `EUtranCellTDD.pmPdcchCceAggregationDlDistr`
- `EUtranCellFDD.pmPdcchCceAggregationUlDistr`
- `EUtranCellTDD.pmPdcchCceAggregationUlDistr`
- `EUtranCellFDD.pmPdcchDlDtxWorseDistr`
- `EUtranCellTDD.pmPdcchDlDtxWorseDistr`
- `EUtranCellFDD.pmPdcchUlDtxWorseDistr`
- `EUtranCellTDD.pmPdcchUlDtxWorseDistr`
- `EUtranCellFDD.pmPucchFailSr`
- `EUtranCellTDD.pmPucchFailSr`
- `EUtranCellFDD.pmUlSinrPdcchDistr`
- `EUtranCellTDD.pmUlSinrPdcchDistr`

Operation

This feature enhancement introduces the following attributes:

- `EUtranCellFDD.decouplePdcchDtxThres`
- `EUtranCellTDD.decouplePdcchDtxThres`

A higher threshold value means that decoupling is less likely to be triggered.

Also, this feature enhancement introduces the following attributes as offsets for stopping downlink PDCCH outer loop adjustment triggered by downlink DTX:

- `EUtranCellFDD.pdcchDlOlaDtxOffset`
- `EUtranCellTDD.pdcchDlOlaDtxOffset`

The lower the offset value is, the easier it is to stop downlink PDCCH outer loop adjustment triggered by downlink DTX.

PDCCH downlink outer loop adjustment is maintained when downlink DTX is received if the value of downlink DTX rate minus uplink DTX rate is greater than the sum of the following values:

- `EUtranCellFDD.decouplePdcchDtxThres` and `EUtranCellFDD.pdcchDlOlaDtxOffset` or
- `EUtranCellTDD.decouplePdcchDtxThres` and `EUtranCellTDD.pdcchDlOlaDtxOffset`

This feature enhancement can be disabled through the following settings:

- The value of the `EUtranCellFDD.decouplePdcchDtxThres` or the `EUtranCellTDD.decouplePdcchDtxThres` attribute is set to zero.
- The sum of the following attributes is greater than or equal to `100`:
  - `EUtranCellFDD.pdcchDlOlaDtxOffset` and `EUtranCellFDD.decouplePdcchDtxThres`
  - `EUtranCellTDD.pdcchDlOlaDtxOffset` and `EUtranCellTDD.decouplePdcchDtxThres`

Interfaces

No impact.

Hardware

No specific hardware requirements.

Other Network Elements

It is recommended to use ENM version 24.4 or later to support the MOM changes introduced by this feature. If the Release Independence Manager application is used, the MOM changes are supported on earlier ENM versions also.

Related Information

LTE Decoupled Downlink and Uplink PDCCH Link Adaptation
Enhanced PDCCH Link Adaptation

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